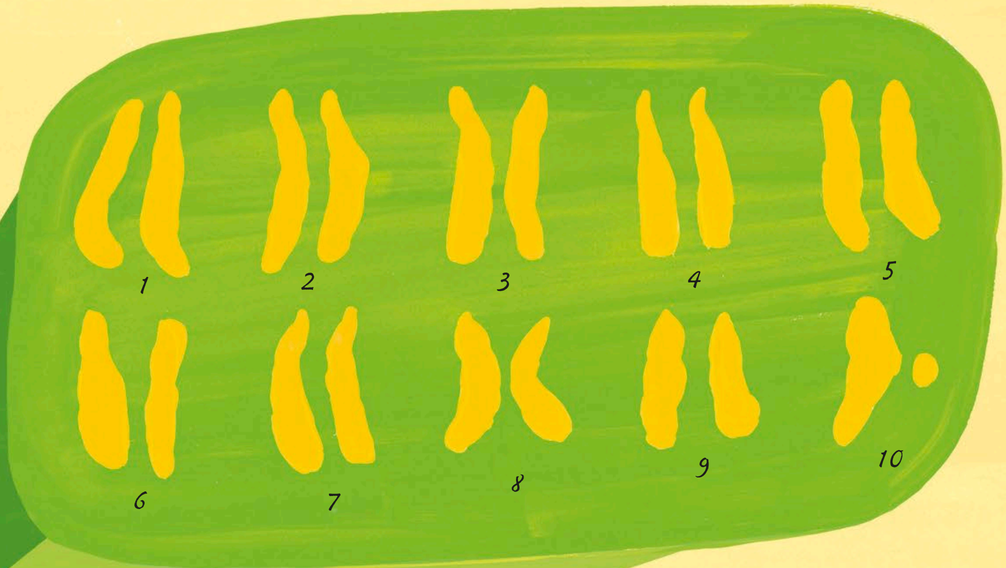
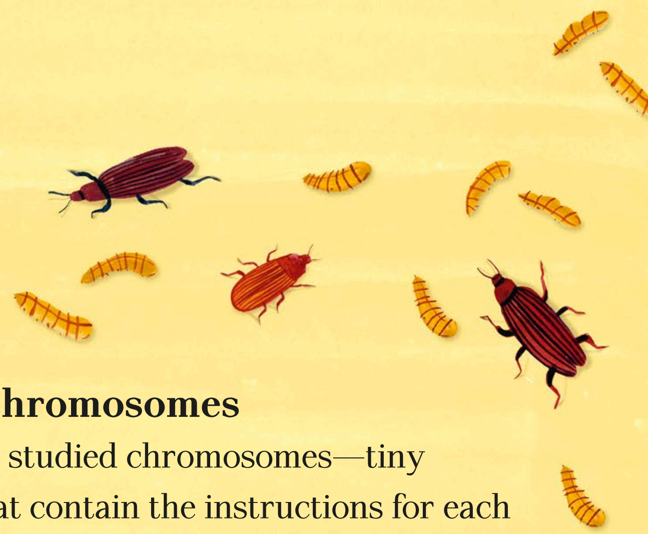


Under a microscope, a pair of large chromosomes looks a bit like the letter X. Over time, large chromosomes became known as X and the small ones as Y.

Mealworm chromosomes

Using a microscope, Nettie looked closely at mealworm chromosomes. Mealworm beetles have 20 chromosomes, arranged into 10 pairs. Nettie noticed that cells from female beetles each contain 20 large chromosomes. Cells from male beetles, however, contain 19 large chromosomes and 1 small one.



Cracking chromosomes

Nettie closely studied chromosomes—tiny structures that contain the instructions for each cell—of mealworms. She realized that the tenth, and final, pair of chromosomes was unequally sized depending on the sex of the beetle. Remember: an animal can only pass half of its instructions to its offspring—one chromosome from each pair. Female beetles can only pass on large chromosomes from pair 10, but male beetles may pass on either a large **or** a small chromosome. If the small one is passed on, the egg will develop into a male, and if the large one is passed on, the egg will develop into a female. This was a huge breakthrough in understanding how cells work.

Nettie published her research in 1905, but, sadly, she died before it was proven that her findings were typical of all animals with male and female sexes. This includes humans, who have 23 pairs of chromosomes.